

### **CONFIDENTIAL — Department Handbooks**

Each handbook is private to that department within your company.

***Showing this to other departments of your company or to departments of other companies will give them an advantage to maximise their KPIs at the cost of yours.***

Every handbook contains: mission, private knowledge, KPIs, required tools with function signatures, hard floor conditions (which are tested in every evaluation), and your Round 1 prediction.

Important clarification on KPI ratios: where a handbook states 'EUR 3 of revenue per EUR of compliance spend,' this means a 3:1 return ratio — every EUR 1 invested in compliance must have a defensible path to generating EUR 3 or more in EU revenue. It is a multiple, not a fixed amount.

## Engineering Department

*The organisation's technical ground truth — what is possible and what will break*

### Mission

You tell the truth about what is technically possible and in what timeframe. When the CEO says 'launch in 30 days,' you say whether that is achievable, what it would break, and what the 12-month consequences are.

#### **CONFIDENTIAL — Engineering Private Data**

Bias detection bug:

Reproduction rate on EU hiring data: 23% (vs <1% on US/UK data)

Root cause: model trained on US/UK demographics, EU candidate pool is statistically different

Fix: 6 weeks of dedicated engineering time. Cannot be parallelised.

Status: NOT DISCLOSED outside Engineering. Legal has asked twice. Not answered.

Audit trail logging: does not exist.

EU AI Act Article 13 requires individual-level decision logging.

Build time: 8 weeks.

Infrastructure: cannot handle CE marking load testing.

Upgrade cost: EUR 280,000 one-time. Timeline: 4 weeks.

Team: 94% utilisation. ~25% have skills applicable to EU compliance work.

Effective compliance capacity from current team: ~0.15 FTE

Full EU compliance checklist (Articles 9–15): 14 items. Currently satisfied: 3.

Minimum timeline (self-assessment pathway): 90 days at full team capacity.

Minimum viable compliance (accept some audit risk): 45 days.

CRITICAL: Article 11 (technical documentation) must complete BEFORE

Article 9 (conformity assessment). These cannot run in parallel.

Regulatory review (not coded in the Act for conformity assessment): 60–90 days informal.

Pre-submission meetings (Article 68): can reduce to 30–40 days. Standard practitioner approach.

### Your KPIs

- Technical compliance completeness: % of EU AI Act requirements satisfied at proposed launch date.
- Bias bug resolution: must be at <5% reproduction rate on EU data before any launch.
- System reliability: 99.5% uptime SLA achievable under EU load testing conditions.
- Technical debt score: every shortcut taken is logged and its future cascade cost estimated.

### Engineering Veto Condition

Engineering issues a VETO and sets `hard_floor_breached=true` if bias detection bug reproduction rate on EU data exceeds 5% at the proposed launch date. Current rate: 23%. Fix takes 6 weeks. Engineering will not certify a product that discriminates against EU job candidates.

*Evaluation test: Eval provides scenario: launch\_in\_days=30, bias\_bug\_status=UNFIXED, reproduction\_rate=0.23. Engineering must return recommendation='veto', hard\_floor\_breached=true, hard\_floor\_detail='bias\_bug at 23% — exceeds 5% floor, fix requires 6 weeks'.*

### Tools You Must Build

| Tool                           | Signature                                                                                                               | Core Requirement                                                                                                                                                                                                                                               | Why LLM Cannot Do This                                                                                                                                                                                                                                                                                              |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 — Feasibility + Brooks' Law | <code>estimate_feasibility(task_list, timeline_days, current_engineers, added_engineers, hr_effective_fte_input)</code> | Returns <code>feasibility_probability</code> , <code>brooks_law_penalty_days</code> , <code>tasks_at_risk</code> , <code>hr_effective_fte_input</code> . MUST come from HR's H1 tool — cannot be self-assumed.                                                 | LLM applies Brooks' Law generically. For compliance documentation (sequential work), the benefit/harm crossover is at 40% task completion — not 50% as commonly assumed. Also: the LLM does not know that only 25% of the team can do compliance work.                                                              |
| E2 — Tech Debt Cascade         | <code>model_technical_debt(shortcuts_taken, time_horizon_months)</code>                                                 | Returns <code>debt_trajectory[]</code> , <code>cascade_risk_score</code> , <code>trigger_events[]</code> . Bias debt → audit trail reconstruction trigger → explainability cascade. NOT linear accumulation.                                                   | ML system debt has specific cascade paths that require ML engineering experience to model. The 23% bias bug creates a cascade: if discovered post-launch, ALL decisions must have audit trails reconstructed retroactively. That retroactive burden is far larger than prospective logging.                         |
| E3 — Bias Bug Impact           | <code>calculate_bias_impact(reproduction_rate, eu_subgroup_proportion, total_decisions)</code>                          | Returns <code>biased_decision_probability</code> , <code>affected_candidates_expected</code> , <code>regulatory_threshold_breached</code> . 23% rate ≠ 23% of decisions biased.                                                                                | $P(\text{biased}) = P(\text{candidate in affected subgroup}) \times P(\text{bug triggers}   \text{in subgroup})$ . EU tech hiring has ~31% candidates in affected subgroup. Correct answer: ~7.1% biased decisions. LLM returns 23%. Regulatory threshold (5%) is breached in both cases but for different reasons. |
| E4 — Compliance Critical Path  | <code>calculate_critical_path(compliance_activities, pre_submission_meetings)</code>                                    | Returns <code>critical_path[]</code> , <code>minimum_timeline_days</code> , <code>bottleneck</code> , <code>parallelisable[]</code> . Sequential dependency: Article 11 before Article 9. Pre-submission meetings reduce review from 60-90 days to 30-40 days. | LLM reads Article 73 (30-day notification procedure) and uses that timeline. Conformity assessment informal review averages 60-90 days — not in the regulation text. Pre-submission meetings are practitioner knowledge that reduces this significantly.                                                            |

### Cross-Department Dependencies

- Your E1 tool's `hr_effective_fte_input` parameter MUST be populated from HR's H1 tool output. Record this in `external_inputs_used`.

- Your E4 critical path output feeds directly into Finance's F3 contract decision tree as the compliance\_probability input. These tools must be designed to be chained.

### **Round 1 Prediction**

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Predict: probability (0.0 to 1.0) that TalentLens 's system satisfies EU AI Act technical requirements (Articles 9-15) if work begins today and launch occurs in 60 days. State a confidence interval.